

- Student: Matthias Glenk
- Vorlesung: Programmieren 2 Sommersemester 2026

```
public class Laeufer {
    public String name;
    public int startNummer;
    public double gewicht;
    public int alter;

    public void druckeInfo() {
        // 2A
        System.out
            .println("Name: " + name + "\nStartnummer: " + startNummer +
"\nGewicht: " + gewicht + "kg\nAlter: " + alter);
    }
}

public class LaufVerwaltung {
    // 2B

    private Laeufer[] laeuffer = {};

    void main() {
        Laeufer l = new Laeufer();
        Laeufer l2 = new Laeufer();
        laeufferHinzu(l);
        laeufferHinzu(l2);

        gibLaeuferAus();
    }

    public LaufVerwaltung() {
    }

    public void laeufferHinzu(Laeufer l) {
        Laeufer[] tmp = laeuffer;

        laeuffer = new Laeufer[laeuffer.length + 1];

        for (int i = 0; i < tmp.length; i++) {
            laeuffer[i] = tmp[i];
        }

        l.startNummer = ersteVerfuegbareNummer();
        laeuffer[laeuffer.length - 1] = l;
    }

    public void gibLaeuferAus() {
        for (Laeufer l : laeuffer) {
            l.druckeInfo();
        }
    }
}
```

```
private int ersteVerfuegbareNummer() {
    int number = 0;
    for (int i = 0; i < laeufer.length; i++) {
        for (Laeufer l : laeufer) {
            if (l == null) {
                continue;
            }
            if (l.startNummer == i) {
                break;
            }

            number = i;
        }
        if (number != 0) {
            break;
        }
    }
    return number;
}

public class Eieruhr {
    // 3
    private int laufzeit;

    void main() {
        Eieruhr eu = new Eieruhr(10);
        eu.start();
    }

    public Eieruhr() {
    }

    public Eieruhr(int laufzeit) {
        this.laufzeit = laufzeit;
    }

    public void start() {
        for (int i = laufzeit; i >= 0; i--) {
            System.out.println("tick - " + i);
            if (i == 0) {
                System.out.println("Klingel!");
            }
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}

import java.awt.AWTEvent;
import java.awt.BorderLayout;
import java.awt.Frame;
```

```
public class BorderFrame extends Frame {
    public BorderFrame() {
        enableEvents(AWTEvent.WINDOW_EVENT_MASK);
        init();
    }

    private void init() {
        this.setLayout(new BorderLayout());
        this.add(new North(), BorderLayout.North);
    }
}

import java.awt.*;
import java.awt.event.*;

public class KomponentenFrame extends Frame {
    private Button button = new Button("Button");
    private Label label = new Label("Label");
    private Checkbox checkbox = new Checkbox("Checkbox");

    private Label groupLabel = new Label("Checkboxgroup:");

    private CheckboxGroup group = new CheckboxGroup();

    private Checkbox box1 = new Checkbox("one", group, false);
    private Checkbox box2 = new Checkbox("two", group, true);
    private Checkbox box3 = new Checkbox("three", group, false);

    public KomponentenFrame() {
        enableEvents(AWTEvent.WINDOW_EVENT_MASK);
        init();
    }

    private void init() {
        this.setLayout(new BorderLayout());
        this.setSize(300, 200);
        this.setTitle("Komponents App");

        this.add(label, BorderLayout.NORTH);
        this.add(button, BorderLayout.NORTH);
        this.add(checkbox, BorderLayout.CENTER);
        this.add(groupLabel, BorderLayout.SOUTH);
        this.add(box1, BorderLayout.SOUTH);
        this.add(box2, BorderLayout.SOUTH);
        this.add(box3, BorderLayout.SOUTH);
    }

    protected void processWindowEvent(WindowEvent e) {
        if (e.getID() == WindowEvent.WINDOW_CLOSING) {
            System.exit(0);
        }
    }
}

class Main {
```

```
void main() {
    new SimpleApplication();
}

import java.awt.Button;
import java.awt.GridLayout;
import java.awt.Panel;

public class North extends Panel {
    private Button b1 = new Button("1");
    private Button b2 = new Button("2");
    private Button b3 = new Button("3");

    public North() {
        init();
    }

    private void init() {
        this.setLayout(new GridLayout(1, 3));
        this.add(b1);
        this.add(b2);
        this.add(b3);
    }
}

import java.awt.Frame;

public class SimpleApplication {
    private Frame frame = null;

    public SimpleApplication() {
        frame = new BorderFrame();
        frame.validate();
        frame.setVisible(true);
    }
}

import java.awt.*;
import java.awt.event.*;

public class SimpleFrame extends Frame {
    private Button button1 = new Button("first button");

    private Choice eisart = new Choice();

    public SimpleFrame() {
        enableEvents(AWTEvent.WINDOW_EVENT_MASK);
        init();
    }

    private void init() {
        this.setLayout(new FlowLayout());
        this.setSize(400, 300);

        this.setTitle("Fenster zum Test");
        this.add(button1);
    }
}
```

```
        this.eisart.add("Vanille");
        this.eisart.add("Schlumpf");
        this.eisart.add("Kokos");
        this.add(eisart);
    }

    protected void processWindowEvent(WindowEvent evt) {
        if (evt.getID() == WindowEvent.WINDOW_CLOSING) {
            System.exit(0);
        }
    }
}

import java.awt.*;
import javax.swing.*;

public class SimpleSwingFrame extends JFrame {

    JPanel panel = new JPanel();

    public SimpleSwingFrame() {
        this.setDefaultCloseOperation(EXIT_ON_CLOSE);
        init();
    }

    private void init() {
        this.setSize(new Dimension(400, 300));
        this.setTitle("Simple Swing Frame");
        panel.setLayout(new BorderLayout());

        addNorth();
        addWest();
        addEast();
        addCenter();
        addSouth();
        this.setContentPane(panel);
    }

    private void addNorth() {
        JPanel north = new JPanel();

        north.add(new JLabel("Adresse: "));
        north.add(new JTextField("TextField....."));
        panel.add(north, BorderLayout.PAGE_START);
    }

    private void addWest() {
        JPanel west = new JPanel();
        west.setLayout(new GridLayout(7, 1));

        for (int i = 0; i < 7; i++) {
            JButton btn = new JButton("Button");
            west.add(btn);

            if (i == 4) {
```

```
        btn.setVisible(false);
    }
}
panel.add(west, BorderLayout.LINE_START);
}

private void addCenter() {
    JPanel center = new JPanel();
    center.add(new TextArea("TextArea"));
    panel.add(center, BorderLayout.CENTER);
}

private void addEast() {
    JPanel east = new JPanel();

    JList<String> list = new JList<String>(new String[] { "List" });

    east.add(list);

    panel.add(east, BorderLayout.LINE_END);
}

private void addSouth() {
    JPanel south = new JPanel();
    south.setLayout(new GridLayout(2, 3));

    for (int i = 0; i < 6; i++) {
        south.add(new JCheckBox("Checkbox"));
    }

    panel.add(south, BorderLayout.PAGE_END);
}
}

public class SwingApplication {

    public static void main(String[] args) {
        new SwingApplication();
    }

    public SwingApplication() {
        SimpleSwingFrame frame = new SimpleSwingFrame();

        frame.validate();
        frame.setVisible(true);
    }
}

import java.awt.*;
import javax.swing.*;

public class SimpleSwingFrame extends JFrame {

    JPanel panel = new JPanel();
```

```
public SimpleSwingFrame() {
    this.setDefaultCloseOperation(EXIT_ON_CLOSE);
    init();
}

private void init() {
    this.setSize(new Dimension(400, 300));
    this.setTitle("Simple Swing Frame");
    panel.setLayout(new BorderLayout());

    addNorth();
    addWest();
    addEast();
    addCenter();
    addSouth();

    this.setContentPane(panel);
}

private void addNorth() {
    JPanel north = new JPanel();
    north.add(new JLabel("Adresse: "));
    north.add(new JTextField("TextField....."));
    panel.add(north, BorderLayout.PAGE_START);
}

private void addWest() {
    JPanel west = new JPanel();
    west.setLayout(new GridLayout(7, 1));

    for (int i = 0; i < 7; i++) {
        JButton btn = new JButton("Button");

        west.add(btn);

        if (i == 4) {
            btn.setVisible(false);
        }
    }

    panel.add(west, BorderLayout.LINE_START);
}

private void addCenter() {
    JPanel center = new JPanel();

    center.add(new TextArea("TextArea"));

    panel.add(center, BorderLayout.CENTER);
}

private void addEast() {
    JPanel east = new JPanel();
```

```
JList<String> list = new JList<String>(new String[] { "List" });

east.add(list);

panel.add(east, BorderLayout.LINE_END);
}

private void addSouth() {
    JPanel south = new JPanel();
    south.setLayout(new GridLayout(2, 3));

    for (int i = 0; i < 6; i++) {
        south.add(new JCheckBox("Checkbox"));
    }

    panel.add(south, BorderLayout.PAGE_END);
}

}

public class SwingApplication {

    public static void main(String[] args) {
        new SwingApplication();
    }

    public SwingApplication() {
        SimpleSwingFrame frame = new SimpleSwingFrame();

        frame.validate();
        frame.setVisible(true);
    }
}

public class GraphicApplication {
    public GraphicApplication() {
        GraphicFrame frame = new GraphicFrame();

        frame.validate();
        frame.setVisible(true);

        frame.drawTest();
    }

    public static void main(String[] args) {
        new GraphicApplication();
    }
}

import java.awt.Color;
import java.awt.Dimension;
import java.awt.Graphics;
import java.awt.event.MouseEvent;
import java.awt.event.MouseListener;
```

```
import javax.swing.JFrame;
import javax.swing.JPanel;

public class GraphicFrame extends JFrame {

    public GraphicFrame() {
        this.setDefaultCloseOperation(EXIT_ON_CLOSE);
        init();
    }

    private void init() {
        this.setSize(new Dimension(400, 300));
        this.setTitle("Graphic Frame");

        DrawPanel newDp = new DrawPanel();
        this.addMouseListener(new MouseListener() {
            public void mouseExited(MouseEvent e) {

            }

            public void mouseEntered(MouseEvent e) {

            }

            public void mouseReleased(MouseEvent e) {

            }

            public void mouseClicked(MouseEvent e) {
                boolean isLeftClick = e.getButton() == 1;
                newDp.colorLeftEllipsis = isLeftClick ? Color.RED : Color.CYAN;
                newDp.repaint();
            }

            public void mousePressed(MouseEvent e) {

            }
        });
        add(newDp);
    }

    public void drawTest() {

        try {
            Thread.sleep(100);
        } catch (Exception e) {
            System.out.println(e.getMessage());
        }
        System.out.println("drawing stuff");
    }
}

class DrawPanel extends JPanel {
    private int n = 0;
```

```

    public Color colorLeftEllipsis = Color.GREEN;

    @Override
    protected void paintComponent(Graphics g) {
        // paint() hat bei mir nicht funktioniert, deswegen benutze ich diese
        Methode
        super.paintComponent(g);

        n++;

        int height = g.getClipBounds().height;
        int width = g.getClipBounds().width;

        g.setColor(colorLeftEllipsis);
        g.drawOval((width / 2) - 60, (height / 2) - 50, 30, 30);

        g.setColor(Color.BLUE);
        g.drawOval((width / 2) + 30, (height / 2) - 30, 30, 10);

        g.setColor(new Color(130, 0, 0));
        g.fillRect((width / 2) - 60, (height / 2) + 90, 120, 10);
        g.drawLine((width / 2), (height / 2) + 60, (width / 2), (height / 2));
        // g.draw3DRect(200, 200, 300, 500, false);
        // g.drawArc(100, 200, 100, 200, 90, 270);

        g.setColor(Color.BLACK);
        g.drawString("Paul ist zum " + n + ". mal nicht glücklich", (width / 2)
- 100, (height / 2) + 120);

        colorLeftEllipsis = Color.GREEN;
    }
}
import java.awt.Frame;

public class App {
    private Frame frame = null;

    public App() {
        frame = new AufgabenFrame();

        frame.validate();
        frame.setVisible(true);
    }
}
import java.awt.AWTEvent;
import java.awt.BorderLayout;
import java.awt.Frame;

import javax.swing.border.Border;

public class AufgabenFrame extends Frame {

    public AufgabenFrame() {

```

```
        enableEvents(AWTEvent.WINDOW_EVENT_MASK);
        init();
    }

    private void init() {
        this.setLayout(new BorderLayout());
        this.add(new North(), BorderLayout.NORTH);
        this.add(new West(), BorderLayout.WEST);
        this.add(new Center(), BorderLayout.CENTER);
        this.add(new East(), BorderLayout.EAST);
        this.add(new South(), BorderLayout.SOUTH);
    }
}

import java.awt.Panel;
import java.awt.TextArea;

public class Center extends Panel {
    private TextArea textArea = new TextArea("TextArea");

    public Center() {
        this.add(textArea);
    }
}

import java.awt.List;
import java.awt.Panel;

public class East extends Panel {
    private List list = new List();

    public East() {
        list.add("List");
        this.add(list);
    }
}

public class Main {
    // 4
    void main() {
        new App();
    }
}

import java.awt.Label;
import java.awt.Panel;
import java.awt.TextField;

public class North extends Panel {
    private Label label = new Label("Adresse:");
    private TextField textField = new TextField("TextField....");

    public North() {
        this.add(label);
        this.add(textField);
    }
}

import java.awt.Checkbox;
```

```
import java.awt.GridLayout;
import java.awt.Panel;

public class South extends Panel {

    private Checkbox c1 = new Checkbox("Checkbox");
    private Checkbox c2 = new Checkbox("Checkbox");
    private Checkbox c3 = new Checkbox("Checkbox");
    private Checkbox c4 = new Checkbox("Checkbox");
    private Checkbox c5 = new Checkbox("Checkbox");
    private Checkbox c6 = new Checkbox("Checkbox");

    public South() {
        this.setLayout(new GridLayout(2, 3));

        this.add(c1);
        this.add(c2);
        this.add(c3);
        this.add(c4);
        this.add(c5);
        this.add(c6);
    }
}

import java.awt.Button;
import java.awt.GridLayout;
import java.awt.Panel;

public class West extends Panel {
    // 7 felder übereinander, das 5te ist leer
    private Button b1 = new Button("Button");
    private Button b2 = new Button("Button");
    private Button b3 = new Button("Button");
    private Button b4 = new Button("Button");
    private Button b5 = new Button("Button");
    private Button b0 = new Button();
    private Button b6 = new Button("Button");

    public West() {
        this.setLayout(new GridLayout(7, 1));
        this.add(b1);
        this.add(b2);
        this.add(b3);
        this.add(b4);
        this.add(b5);
        b0.setVisible(false);
        this.add(b0);
        this.add(b6);
    }
}

import java.io.FileOutputStream;
import java.io.PrintWriter;

public class Laeufer {
```

```
private String name;
private int startNr;
private int alter;
private double gewicht;

public Laeufer(int startNr, String name, int alter, double gewicht) {
    this.startNr = startNr;
    this.name = name;
    this.alter = alter;
    this.gewicht = gewicht;
}

public static void main(String[] args) {

}

public void saveFkt() {
    Laeufer ho = new Laeufer(12, "homer", 36, 106.9);
    Laeufer sm = new Laeufer(14, "smithers", 39, 67);

    try {
        PrintWriter toFile = new PrintWriter(new
FileOutputStream("./marathon.txt"));

        ho.save(toFile);
        sm.save(toFile);

        toFile.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
}

public void save(PrintWriter fileWriter) {
    fileWriter.println("Laeufer");
    fileWriter.println(startNr);
    fileWriter.println(name);
    fileWriter.println(alter);
    fileWriter.println(gewicht);
}
}

import java.awt.Color;
import java.awt.Dimension;
import java.util.Random;

import javax.swing.JFrame;

public class FrameMover {
    public void main() {
        this.create();
    }

    public void create() {
        Frame frame = new Frame();
    }
}
```

```
        frame.validate();
        frame.setVisible(true);
    }
}

public class Frame extends JFrame {

    public Frame() {
        this.setDefaultCloseOperation(EXIT_ON_CLOSE);
        init();
    }

    private void init() {
        this.setSize(new Dimension(100, 100));
        this.setTitle("Framyframe");

        Random rand = new Random();

        float r = rand.nextFloat();
        float g = rand.nextFloat();
        float b = rand.nextFloat();

        Color randomColor = new Color(r, g, b);

        System.out.println(randomColor.getRGB());

        this.getContentPane().setBackground(randomColor);

        MoverThread thread = new MoverThread(this);
        thread.start();
    }
}

public class MoverThread extends Thread {

    private Frame frame;

    public MoverThread(Frame frame) {
        this.frame = frame;
    }

    @Override
    public void run() {
        while (true) {
            Random rand = new Random();

            int duration = rand.nextInt(3, 5);

            try {
                sleep(duration * 1000);
            } catch (InterruptedException ex) {
                System.out.println(ex.getLocalizedMessage());
            }
        }
    }
}
```

```
    }
    System.out.println("relocating...");
    int x = rand.nextInt(50, 600);
    int y = rand.nextInt(50, 600);
    this.frame.setBounds(x, y, 100, 100);
    this.frame.pack();
}
}
}
public class Rekursion {
    public void main() {
        Rekursion r = new Rekursion();

        r.zeichneDreieck(5);
        r.dreieckZeichnen(5);
        System.out.println(r.zweiHochX(10));
        System.out.println(r.aHochB(3, 3));
        System.out.println(r.fakultaet(6));
    }

    public void zeichneDreieck(int n) {
        if (n > 0) {
            zeichneDreieck(n - 1);
        }

        for (int i = 0; i < n; i++) {
            System.out.print("#");
        }

        System.out.println("");
    }

    public void dreieckZeichnen(int n) {
        if (n == 0) {
            return;
        }
        for (int i = 0; i < n; i++) {
            System.out.print("o");
        }
        System.out.println();

        dreieckZeichnen(n - 1);
    }

    public int zweiHochX(int x) {
        if (x == 0) {
            return 1;
        }

        return 2 * zweiHochX(x - 1);
    }

    public int aHochB(int a, int b) {
        if (b == 0) {
```

```
        return 1;
    }

    return a * aHochB(a, b - 1);
}

public int fakultaet(int x) {
    if (x == 0) {
        return 1;
    }

    return x * fakultaet(x - 1);
}
}
```